PH504M Lab 14: Applications of FFT

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Instructions

- Open Jupyter Notebook.
- Create a new notebook named `Lab14_YourName.ipynb`.
- Solve all problems in this notebook.
- Use the minimum number of code cells for each problem.
- Clearly separate problems using comments, e.g. `# Problem 1`.
- Numbers in square brackets indicate the marks for problems.
- Get your work checked and corrected by the TAs during the lab.

Problem 1: Signal Denoising using FFT

A noisy discrete signal is defined as:

$$x[n] = \sin(2\pi f_1 n\Delta t) + 0.5\sin(2\pi f_2 n\Delta t) + \text{noise}$$

where $f_1 = 5$ Hz, $f_2 = 20$ Hz, and the noise is Gaussian distributed.

1. Generate the signal for $N = 256$ samples (vary n) with a suitable choice of Δt.
2. Compute the Fourier transform of the signal and plot its magnitude spectrum.
3. Identify the dominant frequency components.
4. Apply a frequency-domain filter by retaining only components near f_1 and f_2, and setting all other frequencies to zero.
5. Perform the inverse Fourier transform to reconstruct the signal which will be a denoised signal.
6. Plot and compare:
   - Original clean signal (without noise)
   - Noisy signal
   - Denoised signal

Use the following NumPy functions:

- `np.random.normal`
- `np.fft.fft`, `np.fft.ifft`
- `np.fft.fftfreq`

Applying a frequency-domain filter means:

- Compute the frequency array using `np.fft.fftfreq`.
- Choose a small bandwidth Δf around each signal frequency f_1 and f_2.
- Retain Fourier components for which the frequency satisfies:

  $$|f - f_1| < \Delta f \quad \text{or} \quad |f - f_2| < \Delta f$$

  and also for the corresponding negative frequencies:

  $$|f + f_1| < \Delta f \quad \text{or} \quad |f + f_2| < \Delta f$$

- Set all other Fourier coefficients to zero.